MAT700 Mathematical Methods for Data Mining - Lecture7

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Meta-Algorithm

Definition (Meta Algorithms:)

Algorithms operating on algorithms

If you were going to make an important decision, you'd probably get the advice of multiple experts instead of trusting one person.

Why should the problems you solve with machine learning be any different?

This is the idea behind a meta-algorithm. Meta-algorithms are a way of combining other algorithms.

AdaBoost is considered by some to be the best-supervised learning algorithm.

Online Learning

So far the learning algorithms we considered assumed that all the training data is available before building a model for predicting labels on unseen data points.

In many modern applications data is available only in a streaming fashion, and one needs to predict labels on the fly.

Online learning proceeds in rounds. At each round a training example is revealed to the learning algorithm, which uses its current model to predict the label. The true label is then revealed to the learner which incurs a loss and updates its model based on the feedback provided.

Halving Algorithm

The **halving algorithm** is conceptually simple, yet it illustrates many of the concepts in online learning.

Suppose we have access to a set of n experts, that is, functions f_i which map from the input space X to the output space $Y = \{\pm\}$.

Furthermore, assume that one of the experts is **consistent**, that is, there exists a $j \in \{1, 2, \ldots, n\}$ such that $f_j(x_t) = y_t$ for $t = 1, 2, \ldots, T$.

The halving algorithm maintains a set C_t of consistent experts at time t. Initially $C_0 = \{1, 2, \ldots, n\}$, and it is updated recursively as

$$C_{t+1} = \{i \in C_t : \ f_i(x_{t+1}) = y_{t+1}\}$$

The prediction on a new data point is computed via a majority vote amongst the consistent experts:

$$\tilde{y}_t = \mathit{majority}(C_t).$$

Theorem

The Halving algorithm makes at most $\log_2(n)$ *mistakes.*

Proof.

Let M denote the total number of mistakes. The halving algorithm makes a mistake at iteration t if at least half the consistent experts C_t predict the wrong label. This in turn implies that

$$|C_{t+1}| \leq \frac{|C_t|}{2} \leq \frac{|C_0|}{2^M} = \frac{n}{2^M}.$$

On the other hand, since one of the experts is consistent it follows that $1 \leq |C_{t+1}|$. Therefore, $2^M \leq n$. Solving for M completes the proof. □

The Multiplicative Weights (MW) method

We now turn to the scenario where none of the experts is consistent.

The Multiplicative Weights (MW) method is a simple idea which has been repeatedly discovered in fields as diverse as *Machine Learning*, *Optimization*, and *Game Theory*.

The weighted majority algorithm

Now we briefly illustrate *the weighted majority algorithm* in a simple and concrete setting. This is known as the ***Prediction from Expert Advice problem***.

The weighted majority algorithm

- Imagine the process of picking good times to invest in a stock.
  For simplicity, assume that there is a single stock of interest, and its daily price movement is modelled as a sequence of binary events: *up/down*.

- Each morning we try to predict whether the price will go up or down that day; if our prediction happens to be wrong we lose a dollar that day, and if it's correct, we lose nothing.

- The stock movements can be *arbitrary* and even *adversarial*.

  To balance out this pessimistic assumption, we assume that while making our predictions, we are allowed to watch the predictions of n "**experts**."

- These experts could be arbitrarily correlated, and they *may or may not know what they are talking about*.

- The algorithm's goal is to limit its cumulative losses (i. e., bad predictions) to roughly the same as the best of these experts.

- At first sight this seems an impossible goal, since it is not known until the end of the sequence who the best expert was, whereas the algorithm is required to make predictions all along.

The first algorithm one thinks of is to compute each day's up/down prediction by going with the majority opinion among the experts that day.

But, this algorithm doesn't work because *a majority of experts may be consistently wrong on every single day*.

The weighted majority algorithm corrects the trivial algorithm. It maintains a *weighting* of the experts.

As time goes on, some experts are seen as making better predictions than others, and the algorithm increases their weight proportionately.

The algorithm's prediction of up/down for each day is computed by going with the opinion of the weighted majority of the experts for that day.

Weighted majority algorithm

Initialization: Fix an $\eta \leq \frac{1}{2}$. With each expert i, associate the weight $w_i^{(1)} := 1$.

For $t = 1, 2, \ldots$:

1. Make the prediction that is the weighted majority of the experts' predictions based on the weights $w_1^{(t)}, \ldots, w_n^{(t)}$. That is, predict "up" or "down" depending on which prediction has a higher total weight of experts advising it (breaking ties arbitrarily).

2. For every expert i who predicts wrongly, decrease his weight for the next round by multiplying it by a factor of $(1 - \eta)$:

$$w_i^{(t+1)} = (1 - \eta) w_i^{(t)} \quad \text{(update rule)}.$$

Theorem

After T steps, let $m_i^{(T)}$ be the number of mistakes of expert i and $M(T)$ be the number of mistakes our algorithm has made. Then we have the following bound for every i:

$$M^{(T)} \leqslant 2(1+\eta)m_i^{(T)} + \frac{2\ln n}{\eta}.$$

In particular, this holds for i which is the best expert, i. e., having the least $m_i^{(T)}$.

When $m_i^{(T)} >> 2/(\eta)\ln(n)$ we see that the number of mistakes made by the algorithm is bounded from above by roughly $2(1+\eta)m_i^{(T)}$, i. e., approximately twice the number of mistakes made by *the best expert*. This is tight for any deterministic algorithm.

However, the factor of 2 can be removed by substituting the above deterministic algorithm by a randomized algorithm that predicts according to the majority opinion with probability proportional to its weight.

Proof

A simple induction shows that $w_i^{(t+1)} = (1-\eta)^{m_i^{(t)}}$.

Let $\Phi^{(t)} = \sum_i w_i^{(t)}$ ("the potential function"). Thus $\Phi(1) = n$.

Each time we make a mistake, the weighted majority of experts also made a mistake, so at least half the total weight decreases by a factor $1-\eta$.
Thus, the potential function decreases by a factor of at least $(1-\eta/2)$:

$$\Phi^{(t+1)} \leqslant \Phi^{(t)} \left(\frac{1}{2} + \frac{1}{2}(1-\eta) \right) = \Phi^{(t)} (1-\eta/2) .$$

Thus simple induction gives $\Phi^{(T+1)} \leq n(1-\eta/2)^{M^{(T)}}$.

Proof.

Finally, since $\Phi^{(T+1)} \geq w_i^{(T+1)}$ for all i, the claimed bound follows by comparing the above two expressions and using the facts that

$$\ln(1-x) \leqslant -x \quad \text{for } x \geqslant 0$$
$$-\ln(1-x) \leqslant x + x^2 \quad \text{for } x < 1/2.$$

Now:

$$n(1-\eta/2)^{M(T)} \geqslant (1-\eta)^{m_i^{(T)}}$$

$$M(T)\ln(1-\eta/2) \geqslant m_i^{(T)}\ln(1-\eta) - \ln n$$
$$-M(T)\eta/2 \geqslant m_i^{(T)}\ln(1-\eta) - \ln n$$
$$M(T)\eta/2 \leqslant m_i^{(T)}(-\ln(1-\eta)) + \ln n$$
$$M(T)\eta/2 \leqslant m_i^{(T)}(\eta+\eta^2) + \ln n$$
$$M(T) \leqslant 2m_i^{(T)}(1+\eta) + 2\ln n/\eta.$$

